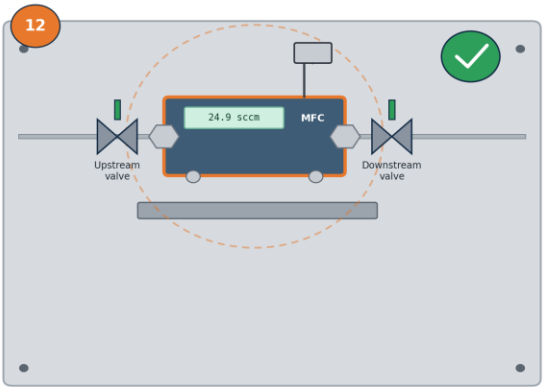


EQUIPMENT MAINTENANCE PROCEDURE

Mass Flow Controller (MFC)

Removal and Installation Procedure

For gas panel modules on semiconductor process tools (CVD / ALD / Etch platforms)



Field	Detail
Document Title	Mass Flow Controller (MFC) Removal and Installation Procedure
Audience	Certified Equipment Maintenance Technicians
Document Standard	Written per Simplified Technical English (STE) and MSTP formatting conventions
Equipment Type	Digital Mass Flow Controller, VCR-fitted gas panel module
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Version	Date	Description	Author
1.0	11-Jul-2026	Initial release.	Sathishkumar S
2.0	15-Jul-2026	Updated the Illustrations.	Sathishkumar S

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1. Introduction

This procedure gives instructions to remove and install a Mass Flow Controller (MFC) on a semiconductor process tool gas panel. An MFC controls and measures the flow of process gas into the process chamber.

Use this procedure to replace a failed MFC. Use this procedure to remove an MFC for calibration or repair.

1.1 Purpose

This procedure gives clear, numbered steps for a safe and correct MFC replacement. Each step includes an illustration of the gas panel.

1.2 Scope

- Removal of a Mass Flow Controller from a gas panel.
- Installation of a Mass Flow Controller on a gas panel.
- Post-installation leak check and functional verification.

1.3 Personnel Requirement

A certified equipment maintenance technician must perform this procedure. The technician must complete facility safety training. The technician must complete Lockout/Tagout (LOTO) training. The technician must complete gas panel hands-on training.

2. Safety Precautions

Read this section before you start the procedure. Follow all safety precautions at all times.

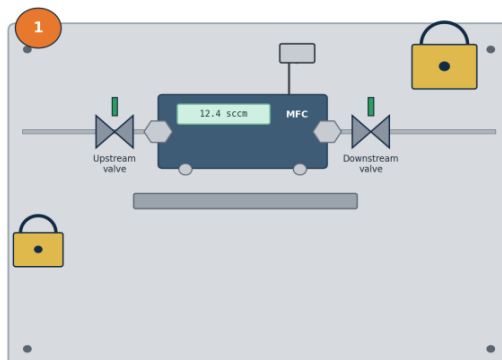


Figure 1. Lock out and tag out the gas supply and electrical power before you start.

WARNING

Process gases can be toxic, flammable, or corrosive. Do not open a gas line before you close the isolation valves and vent the line. Failure to follow this warning can cause serious injury or death.

WARNING

Electrical power presents a shock hazard. Disconnect and lock out the electrical power source before you disconnect the MFC connector.

CAUTION

The MFC body is a static-sensitive device. Wear an ESD wrist strap and connect it to a grounding point before you handle the MFC.

2.1 Required Personal Protective Equipment (PPE)

- Safety glasses
- Chemical-resistant gloves
- ESD wrist strap
- Steel-toe safety shoes

2.2 Lockout / Tagout (LOTO)

1. Identify the gas supply isolation valve and the electrical disconnect for the gas panel.
2. Close the gas supply isolation valve. Apply a lock and a tag.
3. Turn off the electrical disconnect. Apply a lock and a tag.
4. Verify zero energy at the gas panel before you continue.

3. Tools and Materials Required

Item	Specification
Open-end wrench set	1/2 in and 9/16 in, for VCR fitting nuts
Torque wrench	0–100 in-lb range, calibrated
Hex key set	Metric, for MFC mounting bolts
ESD wrist strap	Grounded to facility ground point
New VCR gaskets	Nickel or silver-plated, sized to line diameter
Replacement MFC	Same model and flow range as the removed unit
Helium leak detector or snoop solution	For post-installation leak check
Lint-free wipes and isopropyl alcohol	For cleaning fitting surfaces

NOTE

Use a new VCR gasket for every fitting that you open. Do not reuse a VCR gasket.

4. Before You Begin

Complete the checks in this section before you start the removal procedure.

- Confirm that the process tool is idle and not running a process.
- Confirm that you completed Lockout/Tagout on the gas supply and electrical power. See Section 2.2.
- Confirm that you have the correct replacement MFC. Check the model number and flow range.
- Gather all tools and materials listed in Section 3.
- Put on the required PPE listed in Section 2.1.

5. Removal Procedure

Complete all 6 steps in order. Do not skip a step.

Step 1: Lock Out and Tag Out

1. Lock out and tag out the gas supply and the electrical power. See Section 2.2.

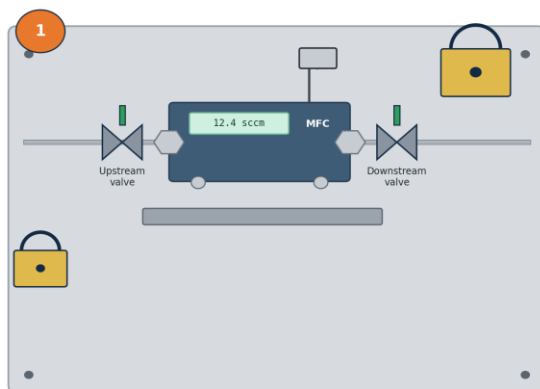


Figure 2. Lock and tag the gas supply and the electrical disconnect.

Step 2: Close the Isolation Valves

2. Close the upstream manual isolation valve.
3. Close the downstream manual isolation valve.

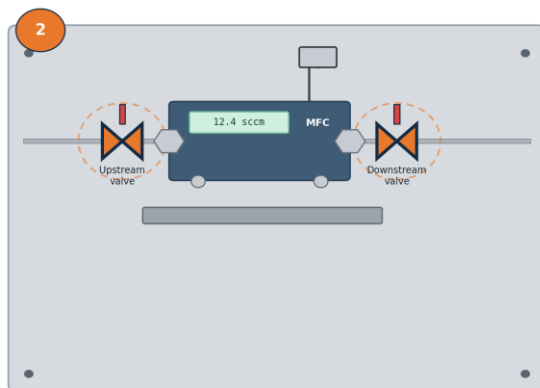


Figure 3. Close the upstream and downstream isolation valves.

Step 3: Vent the Gas Line

4. Vent the gas line between the two isolation valves to the house exhaust.
5. Wait for the line pressure to reach zero. Check the panel pressure gauge.

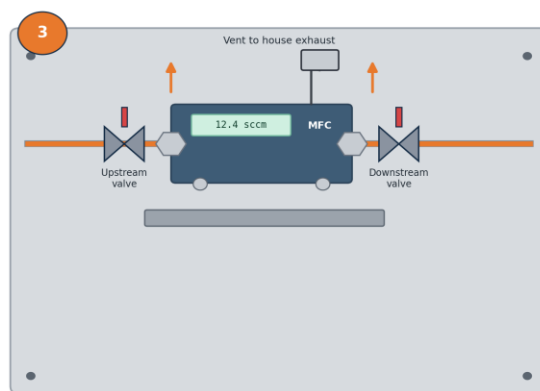


Figure 4. Vent the trapped gas to the house exhaust line.

CAUTION

Do not proceed to Step 4 until the line pressure reads zero.

Step 4: Disconnect the Electrical Connector

6. Disconnect the electrical connector from the top of the MFC.

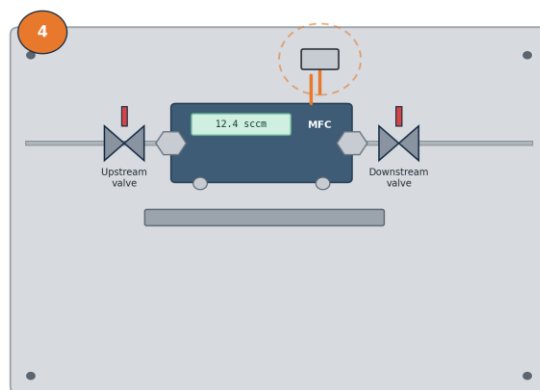


Figure 5. Disconnect the MFC electrical connector.

Step 5: Loosen the VCR Fittings

7. Hold the fitting body steady with one wrench.

8. Turn the fitting nut counterclockwise with a second wrench to loosen the inlet VCR fitting.

9. Repeat this action on the outlet VCR fitting.

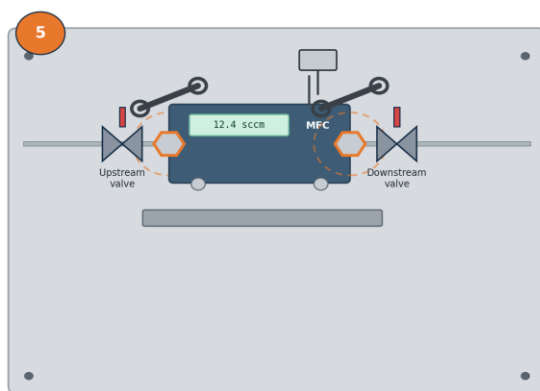


Figure 6. Loosen the inlet and outlet VCR fittings.

NOTE

Support the MFC body with your free hand while you loosen the fittings. This prevents strain on the gas line.

Step 6: Remove the MFC

- 10. Remove the two mounting bolts that secure the MFC to the bracket.
- 11. Lift the MFC straight up and out of the fittings.
- 12. Place the removed MFC in an ESD-safe container.

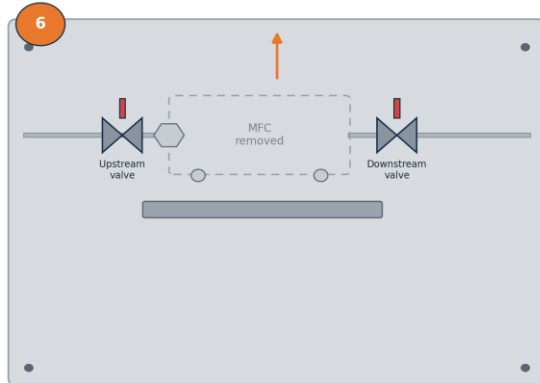


Figure 7. Lift the MFC out of the gas panel.

6. Installation Procedure

Complete all 6 steps in order. Use the replacement MFC identified in Section 4.

Step 1: Position the New MFC

1. Inspect the new VCR gaskets. Confirm that each gasket is clean and free of scratches.
2. Place a new gasket in the inlet fitting and in the outlet fitting.
3. Position the new MFC on the mounting bracket. Align the flow direction arrow on the MFC body with the gas flow direction.

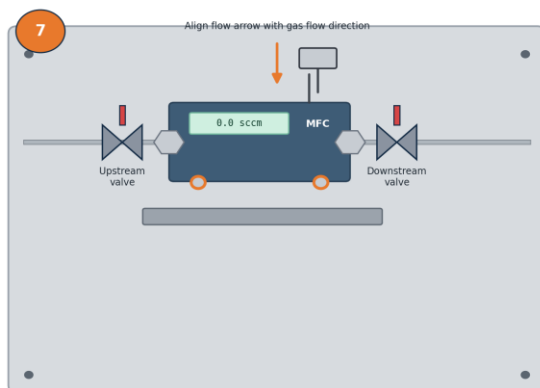


Figure 8. Align the flow direction arrow with the gas flow direction.

Step 2: Secure the Mounting Bolts

4. Install the two mounting bolts. Tighten each bolt by hand.
5. Tighten each mounting bolt to the torque value in Table 1.

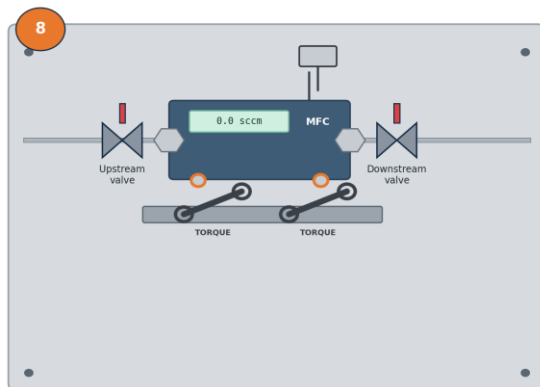


Figure 9. Tighten the mounting bolts to the specified torque.

Step 3: Connect the VCR Fittings

6. Thread the inlet VCR fitting nut by hand until it is finger-tight.
7. Thread the outlet VCR fitting nut by hand until it is finger-tight.
8. Tighten each fitting nut with a calibrated torque wrench to the value in Table 1.

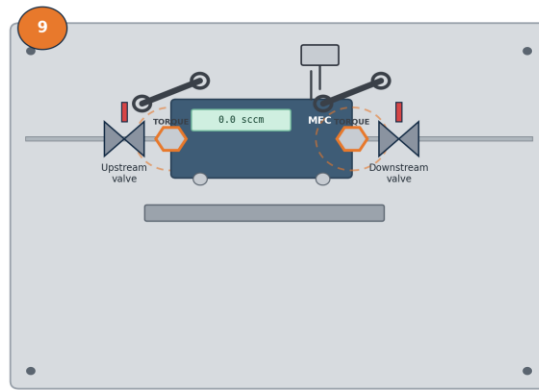


Figure 10. Tighten the VCR fittings to the specified torque.

CAUTION

Do not overtighten a VCR fitting. Overtightening can damage the gasket and cause a leak.

Step 4: Reconnect the Electrical Connector

9. Align the electrical connector with the receptacle on the MFC.

10. Push the connector in until it clicks into place.

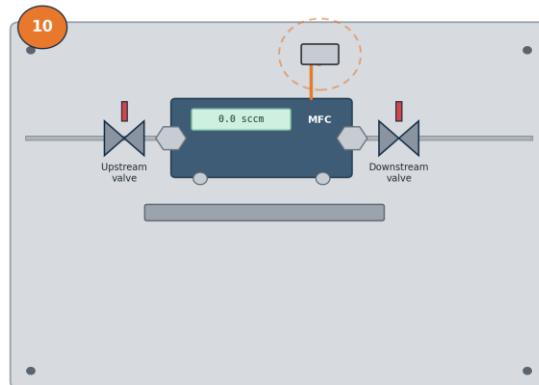


Figure 11. Reconnect the MFC electrical connector.

Step 5: Open the Isolation Valves and Leak Check

11. Remove the lock and tag from the gas supply. See Section 2.2.

12. Open the upstream isolation valve slowly, one quarter turn at a time.

13. Open the downstream isolation valve slowly, one quarter turn at a time.

14. Apply snoop solution, or use a helium leak detector, at the inlet and outlet VCR fittings.

15. Confirm that no bubbles form and that the leak detector reads zero.

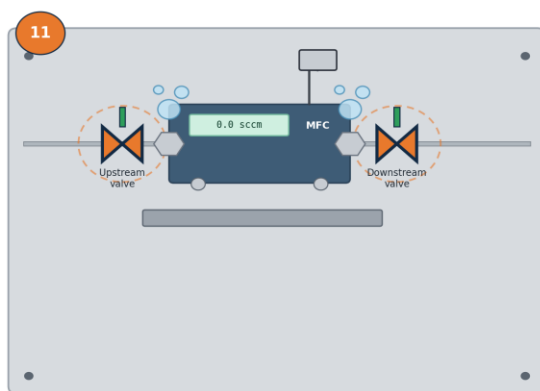


Figure 12. Check both VCR fittings for leaks before you proceed.

WARNING

Do not proceed to Step 6 if you detect a leak. Close both isolation valves. Retighten the fitting to the specified torque. Repeat the leak check.

Step 6: Restore Power and Verify

- 16.** Remove the lock and tag from the electrical disconnect. See Section 2.2.
- 17.** Turn on the electrical power to the gas panel.
- 18.** Confirm that the MFC display shows a stable reading.
- 19.** Run a flow verification test through the tool control system. Confirm that the reading matches the setpoint within the tool specification.

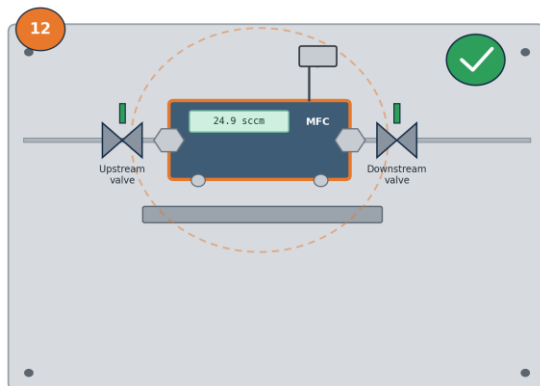


Figure 13. Confirm a stable flow reading on the MFC display.

7. Torque Specifications

Use Table 1 for all torque values in this procedure.

Fastener	Size	Torque Value
VCR fitting nut (1/4 in line)	9/16 in	135–150 in-lb
VCR fitting nut (1/2 in line)	3/4 in	180–200 in-lb
MFC mounting bolt	M4 hex	20–24 in-lb

NOTE

Confirm the torque value against the MFC manufacturer's data sheet. Use the value in this table only if the data sheet is not available.

8. Troubleshooting

Use this table to solve common problems after installation.

Problem	Possible Cause	Solution
The MFC display shows no reading.	The electrical connector is not fully seated, or power is off.	Confirm that the electrical disconnect is on. Reseat the electrical connector.
A leak forms at a VCR fitting.	The gasket is damaged, or the fitting is not at the specified torque.	Close the isolation valves. Replace the gasket. Retighten to the torque value in Table 1.
The flow reading is unstable.	Trapped air in the line, or an upstream pressure fluctuation.	Allow the line to stabilize for 10 minutes. Check the upstream regulator pressure.
The flow reading does not match the setpoint.	Incorrect gas correction factor, or the MFC requires calibration.	Confirm the gas correction factor in the tool control system. Send the MFC for calibration if the reading remains incorrect.
The MFC will not seat on the mounting bracket.	The bracket bolts are misaligned, or the wrong MFC model is used.	Confirm the MFC model and mounting pattern match the removed unit. Realign the bracket bolts.

9. References and Related Documents

- MFC Manufacturer Installation and Operation Manual
- Facility Lockout/Tagout (LOTO) Procedure
- Facility ESD Control Procedure
- Process Tool Gas Panel Schematic Drawing
- Site Environmental, Health, and Safety (EHS) Gas Handling Policy

This procedure is for general reference. Torque values, gasket types, and mounting patterns may vary by MFC model and tool platform. Confirm all values against the manufacturer's data sheet before you perform this procedure.